

Classifying Leukemic B-Lymphoblast Cells from Normal B-Lymphoid Precursors

Using Deep Learning on Microscopic Blood Smear Images

Group 12: Model Citizens

Neil Parikh (Leader), Shaina Banduri, Nishana Dahal
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Outline

- Motivation & Background
- Data Explanation & Acquisition
 - Analysis Pipeline
 - Tricky Analysis Decision
- Bias & Uncertainty Validation
 - Results & Conclusions
 - Next Steps & References

CNNs are a Promising Approach to ALL Diagnosis

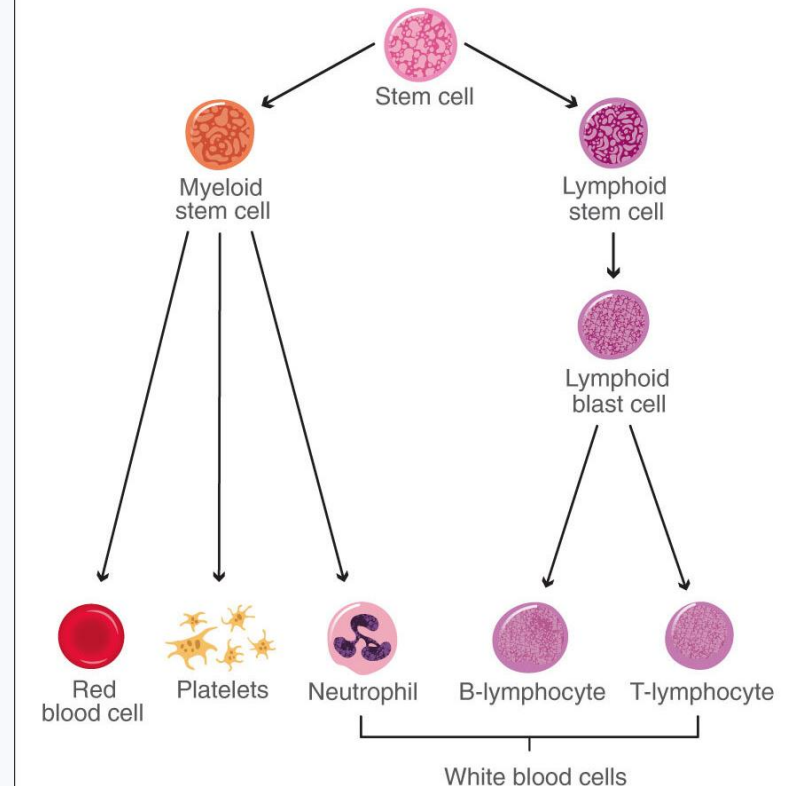
Research Question: Can a fine-tuned CNN reliably classify leukemic B-lymphoblast cells from normal B-lymphoid precursors in microscopic blood smear images?

Why This Matters

- Acute Lymphoblastic Leukemia (ALL) is the most common childhood cancer, accounting for ~25% of pediatric cancers [1].
- Traditional diagnosis requires expert pathologists to manually review blood smear images, and/or expensive laboratory testing
- Automated deep learning classification could accelerate and standardize diagnosis, especially in resource-limited settings

Modeling Approach

- **Transfer Learning:** Fine-tune ResNet50 or VGG16 pretrained on ImageNet
- **Class Imbalance:** Weighted cross-entropy loss + data augmentation
- **Goal:** Weighted F1 score > 0.90 on held-out test set
- **Metric:** Weighted F1 preferred over accuracy due to class imbalance



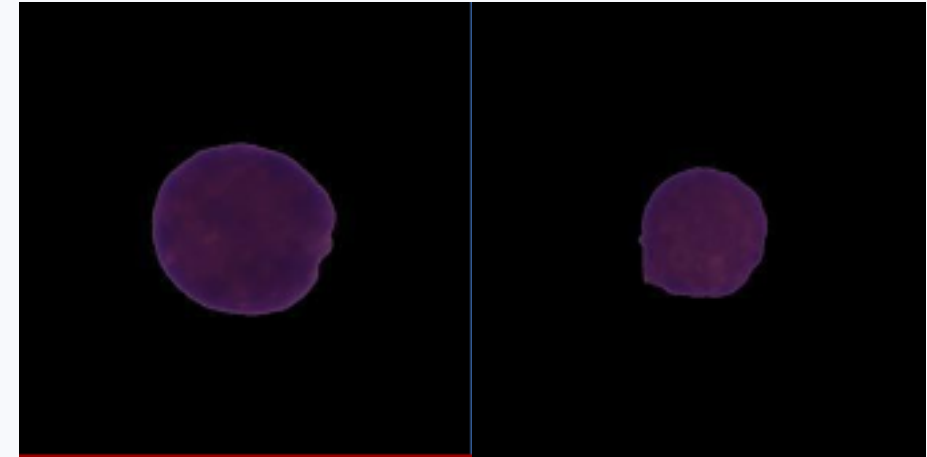
Adapted from Macmillan (2022). [2]

Structure and Background of the C-NMC Dataset

C-NMC 2019 Dataset [3]

Source: The Cancer Imaging Archive (NCI)

- 10,661 microscopic blood smear images
- 73 patients (47 ALL, 26 normal)
- 3 pre-defined cross-validation folds (fold_0, fold_1, fold_2) + held-out test set
- Publicly available; no PII or sensitive patient data



2 examples of C-NMC blood smear images

Challenge 1: Classification of Normal versus Malignant Cells in B-ALL White Blood Cancer Microscopic Images

EDA Reveals Class Imbalance and Brightness Differences in C-NMC

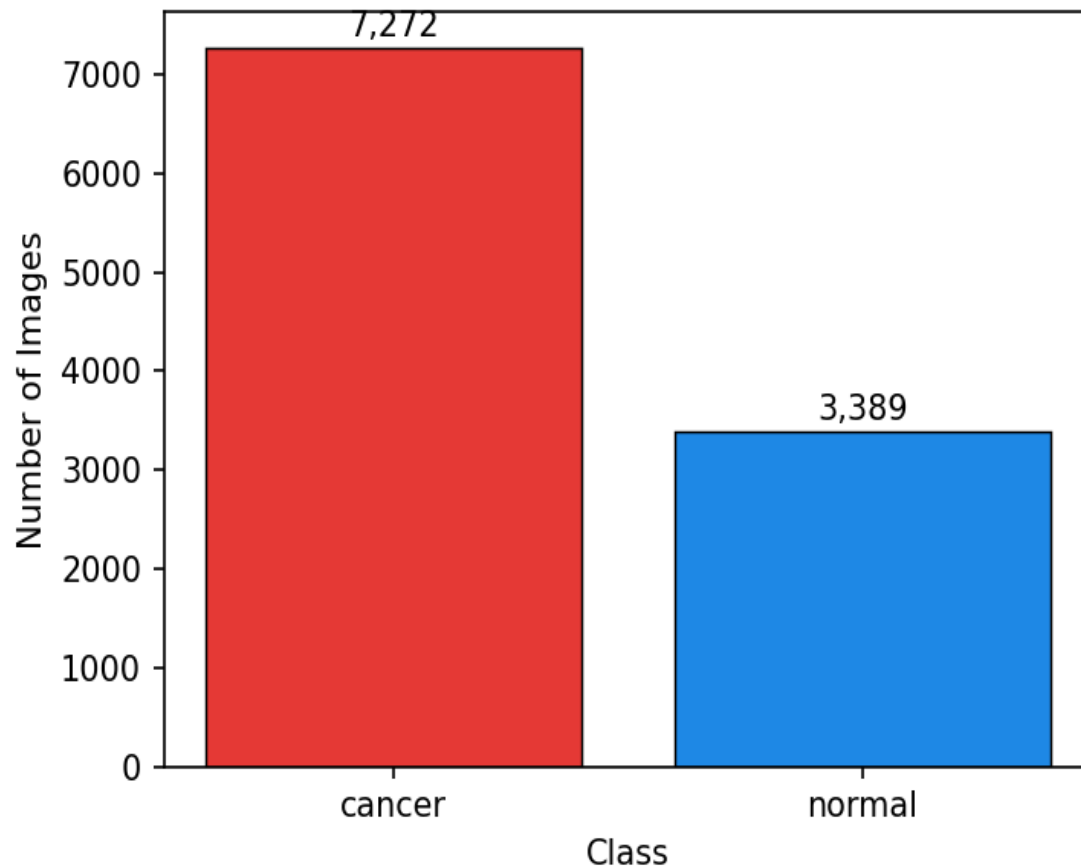
Class Imbalance (~2:1)

- More ALL (leukemia) than normal images
- Requires weighted loss since accuracy alone is misleading

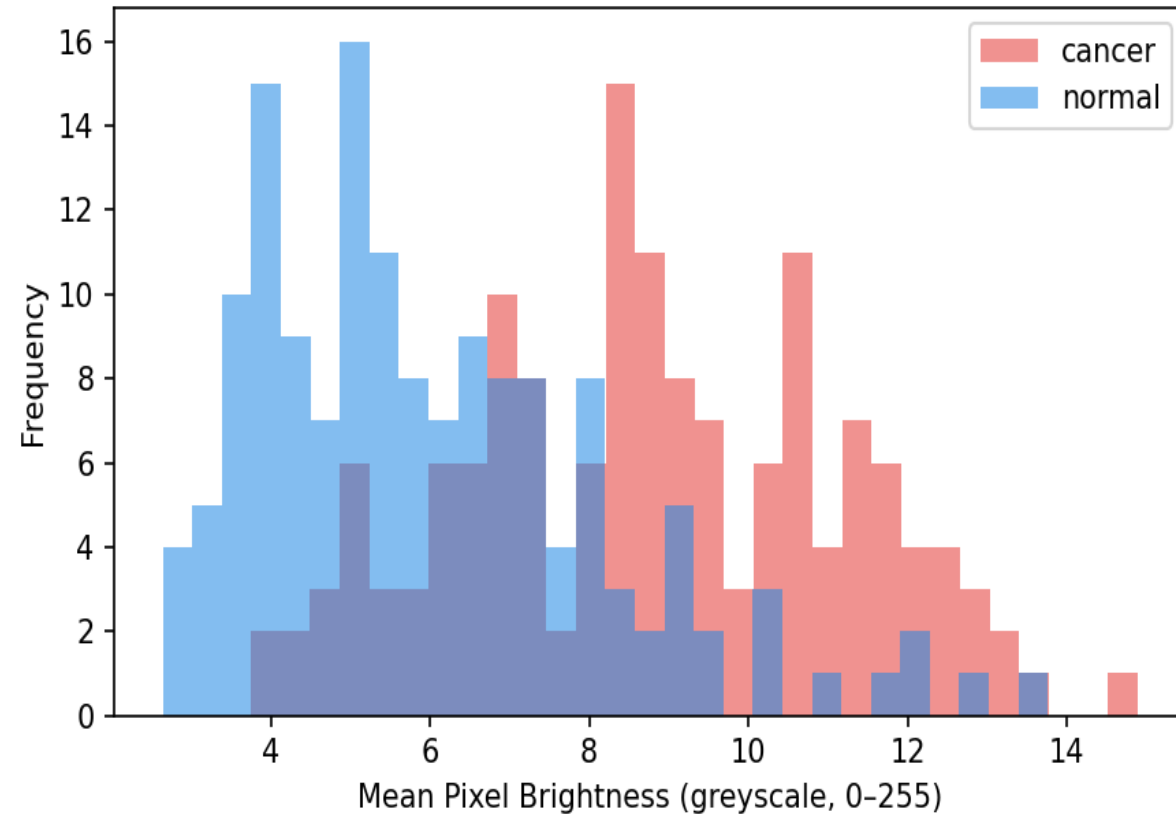
Brightness Difference

- Cancer images: higher avg. brightness, wider spread
- Normal images: concentrated at lower brightness

Class Distribution (Cancer vs Normal)



Average Image Brightness by Class



Analysis Plan

1. Raw Data

10,661 labeled
training images
C-NMC 2019

2. Preprocessing

Resize 224×224
Normalize
(ImageNet
mean/std)

3. Augmentation

Flips, rotations
($\pm 15^\circ$), color jitter
(train only)

4. Model Training

ResNet50 / VGG16
Feature extract
then fine-tune

5. Evaluation

Weighted F1,
Precision, Recall
Confusion Matrix

Data Splitting & Cross-Validation

- fold_0 + fold_1 → train | fold_2 → val (fixed split)
- Patient-aware split: reduces learning of patient-specific patterns
- Held-out test set untouched until final eval

Training Protocol [4]

- Phase 1: Train new head only (feature extraction)
- Phase 2: Unfreeze full network, lower LR (fine-tuning)
- Adam optimizer, up to 30 epochs, early stopping on val. F1

Tricky Analysis Decision

Decision: Choosing Weighted F1 Score as the Primary Evaluation Metric

Why It Was Tricky	Why Not Accuracy?	Impact
• Multiple valid metrics exist (accuracy, precision, recall, F1) • Clinical stakes: false negatives (missed cancer) are far more costly than false positives	• A model predicting ALL for every case would achieve ~68% accuracy but fail entirely on normal cells • Accuracy is misleading when classes are imbalanced	• Weighted F1 penalizes poor performance on both classes, gives a single number accounting for the imbalance • Explicit target: $F1 > 0.90$ serves as our finish line for model selection and final evaluation • We also report precision, recall, confusion matrix for full transparency

Bias & Uncertainty Validation

1. Class Imbalance Bias

- ~2:1 ratio of ALL vs. normal images in training data
- Fix: Weighted F1 more informative than accuracy; confusion matrix reported per class

2. Augmentation Validity

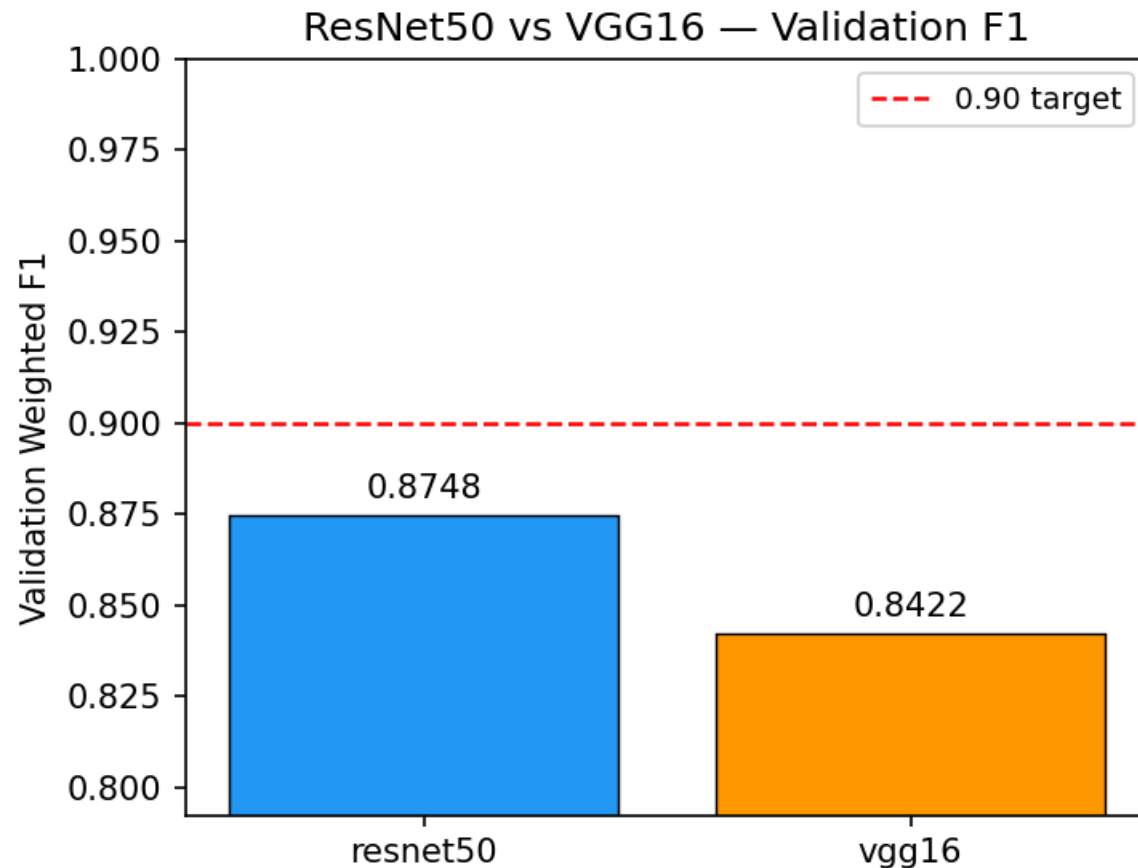
- Augmentation artificially expands training, could introduce unrealistic image variations
- Fix: Only apply to training images; biologically plausible transforms (flips, mild rotation, color jitter)
- Uncertainty: Augmentation strength is a tunable hyperparameter validated on val. F1

3. Unknown Visual Confounders

- Brightness and staining variations may correlate with class label, model may exploit these shortcuts
- Fix: Color jitter augmentation reduces reliance on absolute color/brightness cues
- Results confirm this concern

Results & Conclusions

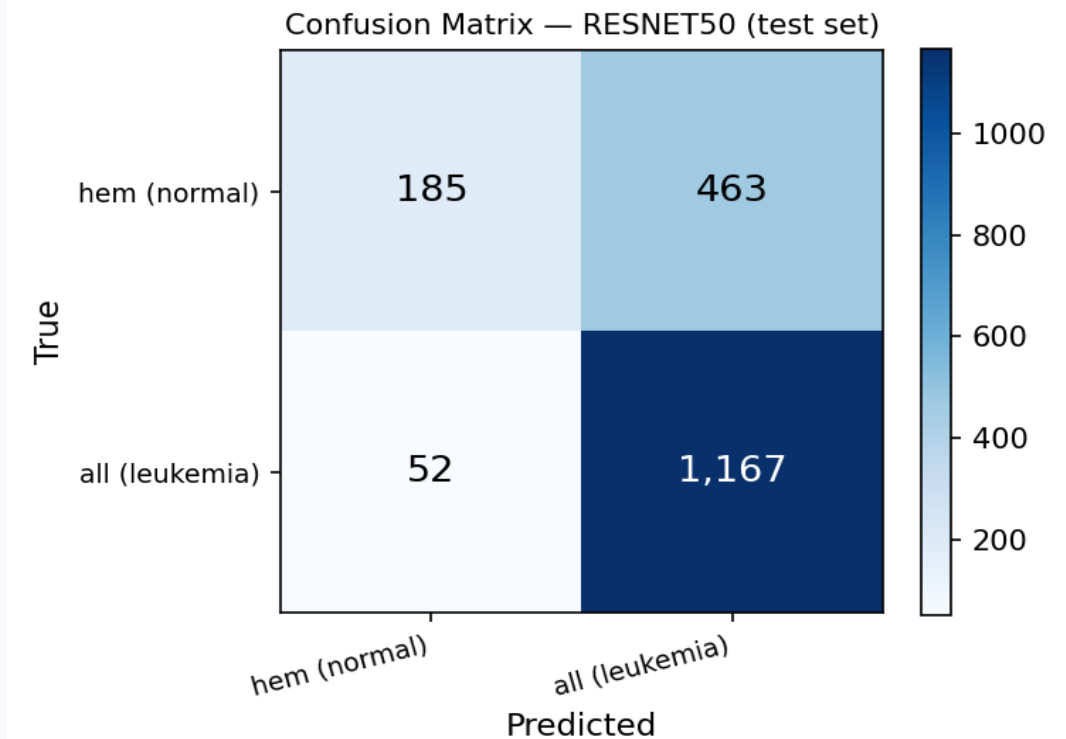
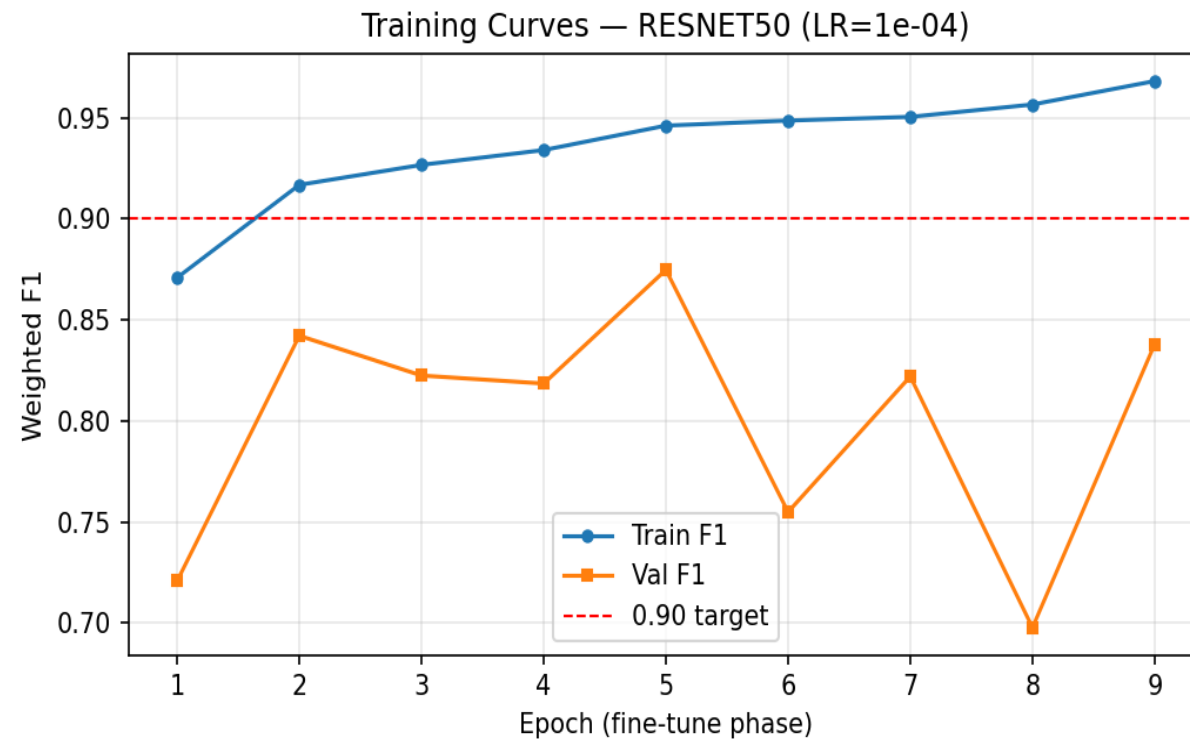
ResNet50 was the best backbone (Val F1 = 0.87), but test fold Weighted did not meet our target (F1 = 0.68 vs 0.90 goal)



Best model: RESNET50 | LR_fine=1e-04 | Batch=32
Val weighted F1 : 0.8748
Test weighted F1: 0.6800



ResNet50 Fine Tuning Performance



- Fine-tuning learning rate of 1e-4, Batch Size: 32
- Weighted F1 Performance drops to 0.68 on the testing set

Next Steps

Improvements	Open Questions
• Address the overfitting trend during fine tuning: • Identifying unexpected features which the model picks up on • Mitigating said identifying features • Grad-CAM integration • Explore improvements to the fine-tuning process to achieve model performance ($F1 > 0.90$) • Ensemble methods	• What patient-level or slide-level features are still driving predictions despite the controls we implemented? If we mitigate that factor... • Can the classifier performance go one step further - ALL subtype detection?

References

- [1] American Cancer Society. "Key Statistics for Childhood Leukemia." 2023.
<https://www.cancer.org/cancer/types/leukemia-in-children/key-statistics.html>
 - [2] *Acute lymphoblastic leukaemia (all)*. Macmillan Cancer Support. (2022, March 1).
<https://www.macmillan.org.uk/cancer-information-and-support/leukaemia/acute-lymphoblastic-leukaemia-all>
 - [3] The Cancer Imaging Archive. "C-NMC-2019." <https://www.cancerimagingarchive.net/collection/c-nmc-2019/>
 - [4] S. J. Pan and Q. Yang, "A survey on transfer learning," IEEE Trans. Knowl. Data Eng., vol. 22, no. 10, pp. 1345–1359, Oct. 2010.
 - [5] Dattani, S. (2025, June 9). *Childhood leukemia: How A deadly cancer became treatable*. Our World in Data. <https://ourworldindata.org/childhood-leukemia-treatment-history>
- GitHub Repository: <https://github.com/shainabanduri/Project-3-DS4002>

Questions?

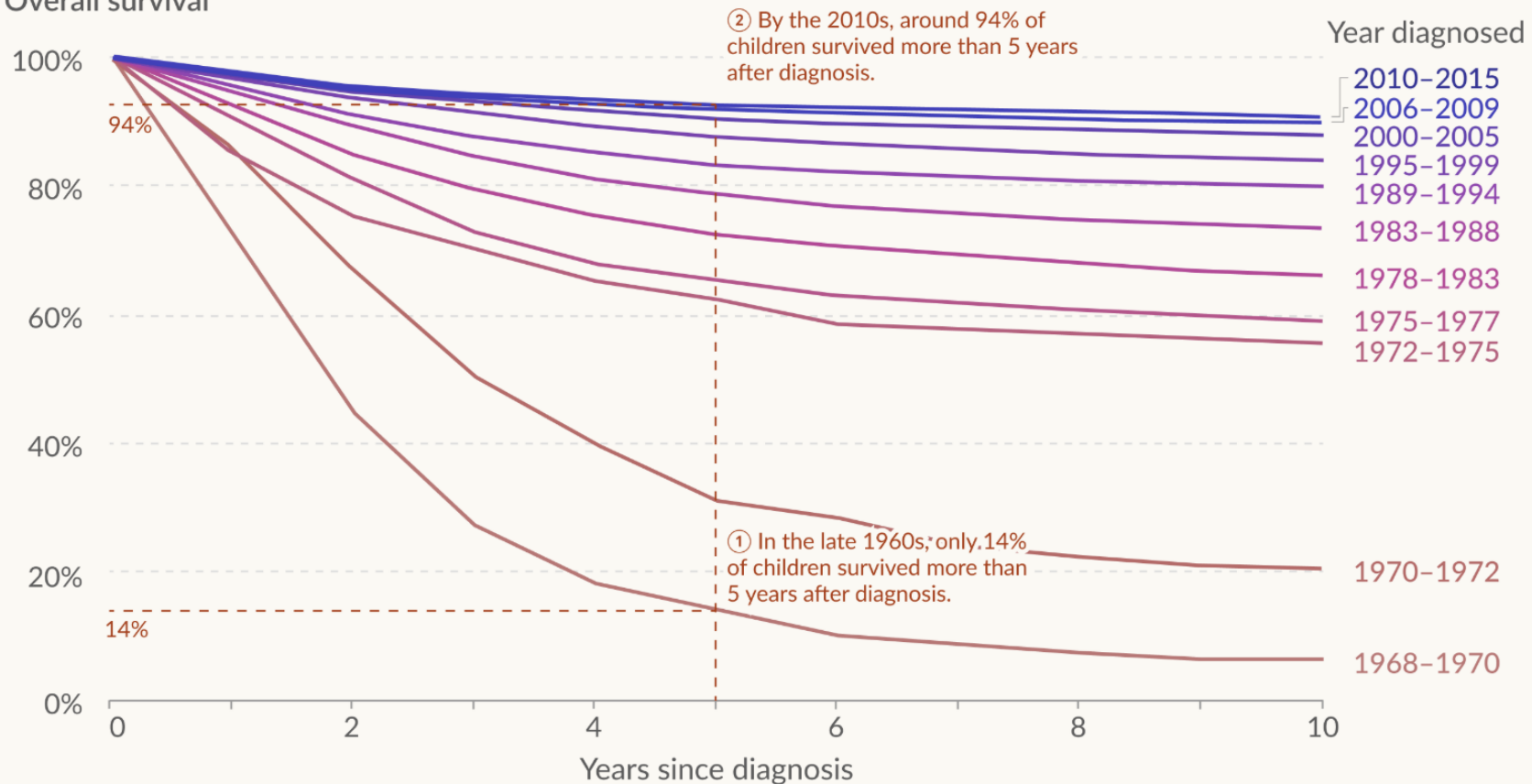
Survival rates for childhood leukemia over time

Our World
in Data

Survival rates in the Children's Oncology Group trials across the United States, Canada, and other countries.

Acute lymphoblastic leukemia (ALL)

Overall survival



Adapted from Our World in Data (2025) [5]